Expert Systems With Applications 195 (2022) 116564

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

Deep reinforcement learning approach for solving joint pricing and inventory problem with reference price effects

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ARTICLE INFO

Keywords:
Dynamic pricing
Inventory control
Reference price effects
Machine learning
Deep reinforcement learning

ABSTRACT

Reference prices, developed by consumers who frequently buy their desired products or services and form psychological price expectations as a benchmark, have significant impacts on customers' purchase behaviors and firms' operational strategies. Therefore, to determine an appropriate pricing and ordering strategy to maximize the total discounted revenues of a retailer, we consider joint pricing and inventory management system under reference price effects in an infinite horizon. Such a system involves uncertain market turbulence and customers' sensitivities to gains and losses (loss-averse, gain-seeking and loss-neutral). We aggregate those factors into a general value function model with only a few realistic constraints on the variables and structure parameters. A deep reinforcement learning approach based on Double Deep Q-Networks with a target network (TN-DDQN) algorithm is proposed and forms the core of the expert decision system. Two ground truth algorithms (value iteration and real-demand-response policy) and two classical RL algorithms (Double Q-learning and Q-learning) are compared with the TN-DDQN algorithm in discrete and continuous state spaces respectively. Through a sequence of experiments, we find that the retailer should not ignore the impact of current prices on future demand, and a myopic policy will cause the retailer's profits to decrease through reference price effects. Moreover, we also find that if customers have high abilities to remember previous prices, the retailer must bring sales prices down and raise the order-up-to level accordingly. Our system with the TN-DDQN algorithm provides a new way to handle complicated behavioral science and operation problems, which can be applied in a broader field.

1. Introduction

Dynamic pricing strategies (also known as yield or real-time pricing strategies) are frequently adopted by companies to balance demand and supply, occupy market shares and increase their long-term profits perpetually. Examples range from electricity (Garcia, Campos-Nañez, & Reitzes, 2005) and urban mobility (Palma, Lindsey, & Proost, 2006; Schröder, Storch, Marszal, & Timme, 2020) to retailing (Boer, 2015) and yield management (Gallego & van Ryzin, 1994; McGill & van Ryzin, 1999). Moreover, appropriate coordination between dynamic pricing and inventory control has a profound impact on firms' service levels and profitability as well as reducing their total costs (Hall, Kopalle, & Krishna, 2003; Elmaghraby & Keskinocak, 2003; Zhu & Thonemann, 2009). For instance, Amazon, Cisco, and Dell dynamically develop everyday-low-price (EDLP) strategies and inventory management systems to adjust inventory levels quickly and achieve higher company profits (Eliashberg & Steinberg, 1987; Chen & Simchi-Levi, 2012; Li & Hou, 2019). Therefore, dynamic pricing and inventory control have a close relationship (Chan, Shen, Simchi-Levi, & Swann, 2004) and the simultaneous determination of these has been adopted by a myriad of manufacturing and retail industries with the help of algorithmic decision systems (Hu, Lu, & Song, 2019; Seele, Dierksmeier, Hofstetter, & Schultz, 2021).

With the development of information network technology and real data-driven management systems, customers have been able to easily observe a commodity's historical price information and companies' dynamic pricing disciplines. Based on the obtained information and repeated transactions, customers have psychological price expectations for their desired products referred to as "reference price" (Kalyanaram & Winer, 1995). Empirical research reveals that the effects of reference price could improve channel efficiency by mitigating the double marginalization effects of price promotion (Lin, 2016) and enhance the sustainable development of supply chains by adjusting the stock levels (Jiang, Huang, & Liu, 2020), finally achieving at least 10 times the expected benefits than a traditional mathematical model without a reference price (Gimpl-Heersink, Rudloff, Fleischmann, & Taudes, 2008).

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https://doi.org/10.1016/j.eswa.2022.116564
Received 20 July 2021; Received in revised form 28 December 2021; Accepted 17 January 2022
Available online 20 January 2022
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Thus, it is necessary to examine the joint pricing and inventory decisions under the circumstance of reference price effects. Although some studies have attempted to identify optimal pricing and inventory control levels with reference price effects via modeling analysis (Chen, Hu, Shum, & Zhang, 2016; Song, Ran, & Shang, 2017), the mathematical models in these studies ordinarily and heavily rely on simplifications and assumptions, which constrains and restricts model application in the real environment. A common reason for model simplification is that traditional expert systems such as dynamic programming (DP) (e.g., value iteration, exact policy iteration, and generalized policy iteration) or heuristics generally do not solve this problem well. The curse of dimensionality plagues DP (Bellman, 1957), resulting in complex and time-consuming computations. Although heuristic algorithms or rule-based policies can find acceptable solutions in reasonable computational time (Beynaghi et al., 2019), they typically require specific circumstances and the problems must satisfy conventional conditions. Further, DP or heuristics requires more knowledge about the environment, which results in less generality and applications for pricing as well as inventory problems with the customer behavior.

In our study, we use deep reinforcement learning (DRL) to cope with the mentioned difficulties without requiring an exact representation of the environment and feature selections. DRL is a combination of deep learning and reinforcement learning (RL) that uses artificial neural networks (ANNs) to formulate the sequential problem as a Markov decision process (MDP). A DRL agent learns to maximize expected value function by interacting with a complicated environment under the exploration versus exploitation trade-off (Elreedy, Atiya, & Shaheen, 2021). Also, DRL provides a framework for model-free learning without simulation models and requires a low computational budget (Alizadeh, Allen, & Mistree, 2020; Dixon, Halperin, & Bilokon, 2020). Along with the rapid development of computational power, artificial intelligence systems have successfully been applied in telerobot systems, scheduling and allocation of resources, health care, and natural language processing (Sutton & Barto, 2018; Zamani Sabzi et al., 2018; Soltanisehat, Alizadeh, Hao, & Choo, 2020). However, applications of DRL as decision making systems in the field of management, particularly in marketing and operation contexts remain scarce. Our goal is to implement a more general value function and model structure in a joint pricing and inventory problem with reference prices and to determine the optimal pricing and ordering policies for the retailer. Further, we formulate an infinite-horizon MDP on the retailer's optimal pricing and inventory control strategies with reference prices and propose a DRL algorithm, based on the Double Deep Q-Network, to solve the intractable problem in behavioral operations.

The main contributions of this paper are summarized as follows:

- To the best of our knowledge, this is novel to unequivocally investigate reference price effects through deep RL methods in operations management.
- A general model with only a few realistic assumptions is constructed that can be practical and flexible in improving the profits of enterprises in a complicated and dynamic market environment.
- A TN-DDQN algorithm is proposed to solve the general model of dynamic pricing and inventory control with reference prices. The experiments show that the TN-DDQN algorithm not only has a near-optimal performance but it also has a robust advantage under marketing environment stochastic disturbances.
- Suggestions are given for retailers (decision makers) on how to explore customers' price perceptions and provide better customer service.

The rest of the paper is organized as follows. We review the relevant literature in **Section 2**. **Section 3** develops the reference prices and value function model. A DRL algorithm is then proposed in **Section 4** to solve the aforementioned mathematical model. In **Section 5**, we conduct several numerical experiments to evaluate the performance of the designed algorithm and demonstrate the impacts of the customer behavior. We end with concluding remarks and future research orientations in **Section 6**.

2. Literature review

We classify prior literature related to this work into two categories: pricing and inventory management with reference price effects, and RL techniques in supply chain management (SCM).

2.1. Pricing and inventory management with reference price effects

The joint consideration of pricing and inventory management has been attracting considerable attention in traditional operations management (e.g., Petruzzi & Dada, 1999; Elmaghraby & Keskinocak, 2003; Chan et al., 2004; Chan, 2019). This study focuses on discovering pricing and inventory control strategies with memory-based reference price effects that are supported by extensive research in psychology (Mazumdar, Raj, & Sinha, 2005).

The early studies in pricing and inventory management with reference price effects are generally based on the newsvendor model, which is a fundamental decision model in operations management. Urban (2008) analyzes a single-period newsvendor model with both symmetric and asymmetric reference price effects, adopting a numerical analysis in which the reference price effects had a significant impact on companies' profits. After that, two-period scenarios are explored (Gimpl-Heersink, 2008; Taudes & Rudloff, 2012) in which the base-stock list-price (BSLP) policy is proven under rigorous constraints on the single-period revenue, inventory cost functions, and demand sensitivities associated with reference price effects. It should be noted that the BSLP policy is first proposed by Porteus (1990), when he develops stochastic inventory theory. Federgruen and Heching (1999) also prove the BSLP policy when dealing with women's apparel retailers' pricing and replenishment decision strategies.

With respect to a multi-period discounted model, two types of transformation techniques are introduced to tackle technical challenges. Güler, Bĭlgiç, and Güllü (2014), Güler et al. (2015) inverse the demand function and transform order-up-to level to safety-stock. They show that the optimal decision is a state-dependent order-up-to (BSLP) policy, under the strict assumption of concave transformed single-period revenue function. The second form of concave transformation approach is introduced by Chen et al. (2016). They assume that the sensitivities of reference price effects and customers' memories must satisfy specific conditions, such as that customers are loss-neutral or they have short memories. Then, the optimality of state-dependent BSLP policy is proven and the correlation between the optimal steady-state price and consumers' preference is basically consistent with Popescu and Wu (2007). The main differences between our study and all of the above literature are that we consider the problem as an infinite horizon MDP without rigorous assumptions of mathematics and explicitly propose specific solutions in our expert systems. Specifically, we have a general value function with no restriction on customers' preferences, and no complicated transformation techniques. Thus, compared with prior studies, this study is more practical and extensible.

2.2. Reinforcement learning techniques in supply chain management

Reinforcement learning techniques, including classic RL and DRL, are an area of machine learning where an agent adapts its policy to maximize its total rewards when interacting with the environment. It has been successfully applied to solve various problems in traditional SCM from a single-product to multiple products, as shown in Table 1.

Taking underlying production and allocation techniques into account, the SCM problem presents itself as a nondeterministic polynomial-time (NP-)-hard algorithmic problem with combinatorial complexity (Stockheim, Schwind, & Koenig, 2003). As a consequence,

Table 1
Reinforcement learning techniques applied to traditional SCM.

Reinforcement Learning Techniques	Products	Existing studies
classic RL		
SMART	a single product	Giannoccaro and Pontrandolfo (2002)
	a single product	Pontrandolfo et al. (2002)
Case-based reasoning	a single product	Kwon et al. (2008)
	a single product	Jiang and Sheng (2009)
Q-learning	a single product	Stockheim et al. (2003)
	a single product	Chaharsooghi et al. (2008)
SARSA	a single product	Kemmer et al. (2018)
	a single product	Kara and Dogan (2018)
DRL		
DQN	a single product	Oroojlooyjadid et al. (2017)
	multiple products	Suetsugu et al. (2019)
	a single product	De Moor et al. (2021)
PPO	a single product	Vanvuchelen et al. (2020)
A2C	multiple products	Meisheri et al. (2020)
	multiple products	Sultana et al. (2020)
A3C	a single product	Gijsbrechts et al. (2020)

the value-iteration based semi-Markov average reward technique (SMART) (Das, Gosavi, Mahadevan, & Marchalleck, 1999) and case-based reasoning technique (Santamaria, Sutton, & Ram, 1997) are drawn from experience and have been successfully applied to linear supply chains (Giannoccaro & Pontrandolfo, 2002), distribution-type supply chains (Kwon, Kim, Jun, & Lee, 2008; Jiang & Sheng, 2009), and distributed production systems (Pontrandolfo, Gosavi, Okogbaa, & Das, 2002). Besides the SMART algorithm and case-based technique, temporal-difference prediction methods, such as state–action–reward–state–action (SARSA) and Q-learning algorithms, have been used to handle linear supply chains (Chaharsooghi, Heydari, & Zegordi, 2008), distributed supply chains (Kemmer, von Kleist, de Rochebouët, Tziortziotis, & Read, 2018), decentralized networks (Stockheim et al., 2003), and single-echelon perishable inventory systems (Kara & Dogan, 2018).

However, classic RL algorithms are only appropriate for a limited state-space size and suffer from a lack of scalability. Therefore, in recent years, RL has been combined with deep learning, where the value and policy functions are approximated by deep neural networks (DNNs). The outstanding performance of the Deep Q-Network (DQN) algorithm in a series of Atari games discovered by Mnih et al. (2015) is a significant milestone, and this is followed by notable extensions such as dueling networks (Wang et al., 2016) and prioritized experience replay (Schaul et al., 2015). Oroojlooyjadid, Nazari, Snyder, and Takáč (2017) and De Moor, Gijsbrechts, and Boute (2021) propose a transfer learning approach to using the DQN algorithm to manage inventory replenishment policies, and Suetsugu, Narusue, and Morikawa (2019) extend branching DQN with reward allocation in a multi-product inventory system. Additionally, the application of policy gradient methods is extended to SCM, especially to joint replenishment problems (Vanvuchelen, Gijsbrechts, & Boute, 2020). As a combination of value-based and policy gradient methods, actor-critic methods have a critic that estimates value functions directly and an actor that updates the policy in the direction suggested by the critic. Recently, actor-critic algorithms have been applied to continuous-time multi-product inventory management (Meisheri, Baniwal, Sultana, Khadilkar, & Ravindran, 2020; Sultana et al., 2020) and inventory problems with three scenarios: lost sales, dual-sourcing, and multi-echelon (Gijsbrechts, Boute, Van Mieghem, & Zhang, 2020).

It is worth mentioning that few studies assume that the demand is endogenously determined and that agents simultaneously make ordering and pricing decisions. This topic is more relevant to our study. Dogan and Güner (2015) concentrate on competitive pricing and inventory management, where customers select the winning retailer in accordance with random utility maximization using a Q-learning-based RL algorithm. Wang, Gan, Li, and Yan (2021) focus on joint pricing and perishable inventory problems with non-zero lead time and the homogeneous Poisson process by means of DRL algorithms. Although, unlike traditional SCM, they take the pricing decision into account, we also bring reference price effects into focus, which integrates behavioral science into traditional operations management with RL techniques.

Reference price effects can affect the mechanisms of pricing on customers' demands, and then change firms' ordering strategies. Therefore, firms need to jointly consider pricing and inventory control under reference price effects in order to maximize their profits. Consequently, in this paper, by using a deep network with felicitous layer structures and feature selection, our Double Deep Q-Network combines the superiorities of deep learning and Double Q-learning and is shown to be valid and suitable for the joint pricing and inventory problem with reference price effects. Further, we use DNNs to circumvent the curse of dimensionality and prove that our DRL algorithm outperforms other RL algorithms in most cases and sheds new light on the combination of decision theory and machine learning.

3. Problem description and modeling

This section introduces the framework of the joint pricing and inventory problem with reference price effects in preparation for the following DRL algorithm design.

3.1. Sketch and notations

Consider a retailer facing a single-product, periodic-review inventory and pricing problem in an infinite planning horizon with discount criterion. At the start of each period t, the retailer observes customers' reference prices, r_t and initial inventory level, x_t, then must simultaneously decide on the order-up-to level, y_t and sales price, p_t. After the demand arrives, the retailer captures the current period's reward, R_t and the inventory level drops, followed by holding or backlogging inventory cost. Fig. 1 illustrates the interactions between the retailer and market environment. The retailer dynamically adjusts the order-up-to level and selling price day-by-day to maximize the expected rewards over the infinite horizon. We summarize the notations of the joint pricing and inventory control with reference prices in Table 2.

3.2. Reference price effects

Based on the adaptation level theory (Helson, 1964), if the current selling price is higher or lower than the reference price, it is perceived as psychological loss or gain, respectively. In terms of their susceptibility to gains and losses, customers are generally classified into one of three categories: gain-seeking customers, who are more susceptible to gain than loss and more responsive to product discounts; loss-averse customers, who are more susceptible to loss than gain and emerge with unique characteristics in prospect theory experiments (Kahneman & Tversky, 1979); and loss-neutral customers, who evaluate loss and gain indistinguishably. Two economic models are widely used for expressing and modeling reference price effects (Popescu & Wu, 2007). The first is the "absolute difference" (AD) model and the reference price effect

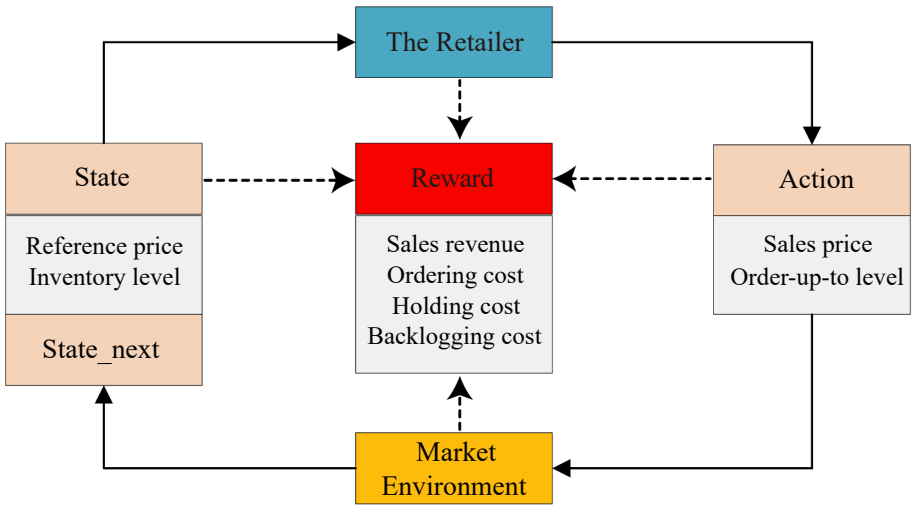

Fig. 1. Interactions of the retailer and market environment.

Table 2
Notations grouped by the elements of model analysis.

Notation	Explanation
p_t	Sales price charged by the retailer at period t
r_t	Customers' reference prices at period t
y_t	Order-up-to inventory level at the beginning of period t
x_t	Initial inventory level at the period t
η^+	Parameters of reference price effects when $r_t \geqslant p_t$
η^-	Parameters of reference price effects when $r_t \leqslant p_t$
α	Memory factors
p_{min}	The lower bound of sales price
p_{max}	The upper bound of sales price
c_t	Unit ordering cost
h	Unit holding cost
v	Unit backlogging cost
σ_t	Random variable in demand function at period t
γ	Discount factor

$R(r-p,r) = h(r-p)s(r)$, with h increasing, $h(0) = 0$, and s is a positive decreasing scaling function. The second is the "relative difference" model, in which the reference price effect $R(r-p,r) = h((r-p)/r)$, with h increasing and $h(0) = 0$. In this paper, we adopt the AD model due to its wide availability (Chen et al., 2016; Fibich, Gavious, & Lowengart, 2003; Kopalle, Rao, & Assunção, 1996; Greenleaf, 1995). Specifically, we define $R(r-p,r) = \eta(r_t - p_t)$ as the customers' reference price effects in period t and it can be written as follows:

$$R(r-p,r) = \eta^+ max\{0, r_t - p_t\} + \eta^- min\{0, r_t - p_t\} = \begin{cases} \eta^+(r_t - p_t), \ if \ r_t \geqslant p_t \\ \eta^-(r_t - p_t), \ if \ r_t \leqslant p_t \end{cases} \tag{1}$$

where the nonnegative parameters η^+ and η_-, judge the customers' sensitivities toward the reference prices and selling prices regarding gain and loss, respectively. Such as, customers are called loss-averse if $\eta^- \geqslant \eta^+$, gain-seeking if $\eta^+ \geqslant \eta^-$, and loss-neutral if $\eta^+ = \eta^-$. The approach accounting for the heterogeneity in customer valuation is proposed in Hu and Nasiry (2018).

The customer's reference price is a weighted average of the observed selling price and the reference price of previous period (Mazumdar et al., 2005). A frequently used and analyzed model for the update of the reference price is the exponentially smoothed adaptive expectations model:

$$r_{t+1} = \alpha r_t + (1-\alpha)p_t \tag{2}$$

where $0 \leqslant \alpha \leqslant 1$ is the memory factor, and the larger the memory factor, the longer the customer's memory. In more detail, the current sales price has a deeper effect than the previous reference price when α is small.

3.3. Value function

In each period, the expected demand d_t, is a nonnegative function of the price p_t, and the reference price effect $R(r_t - p_t, r_t)$. The randomness of demand is modeled by incorporating an additive term over the expected demand d_t, with $E[\sigma_t] = 0$. Hence, the general demand function is given by:

$$d_t(p_t, r_t) = d_t(p_t, R(r_t - p_t, r_t)) + \sigma_t \tag{3}$$

When the random variable is additive to the demand model, the variance of demand is independent of price and the demand coefficient of variation is an increasing function of price (Petruzzi & Dada, 1999).

We assume that linear order cost c_t, must be smaller than the lower bound of sales prices and unmet demand is all backlogged. The inventory cost is formulated as $H_t(z) = hz^+ - vz^-$, where $h(>0)$ is the per period unit holding cost and $v(>0)$ is the per period unit backlogging cost. $z^+ = max\{z, 0\}$ and $z^- = min\{z, 0\}$ are the inventory levels at the end of period t. The expected total inventory cost in period t is given by:

$$G_t(y_t, p_t, r_t) = \mathrm{E}\{H_t(y_t - d_t(p_t, r_t))\} \tag{4}$$

The retailer's objective is to jointly optimize inventory and pricing policy to maximize the total expected discounted profits over the infinite horizon. The profit-to-go value function $V_t(x_t, r_t)$, satisfies the subsequent DP recursion:

$$V_t(x_t, r_t) = \max_{y_t, p_t} \mathrm{E}\{p_t d_t(p_t, r_t) - c_t(y_t - x_t) - G_t(y_t, p_t, r_t) + \gamma \mathrm{E} V_{t+1}(y_t - d_t(p_t, r_t), \alpha r_t + (1-\alpha)p_t)\} \tag{5}$$

where $\gamma \in [0,1]$ is the discount factor, and decision variables y_t and p_t satisfy $y_t \geqslant x_t$ and fall in a feasible price interval $[p_{min}, p_{max}]$ as well as $p_{min} \geqslant c_t$, which guarantees the real environment. The discussed DP problem admits some finite optimal solution and becomes intractable. Moreover, in a sequential decision problem over infinite horizon, all parameters can be restricted to stationary without loss of generality.

4. Deep reinforcement learning model

There are several methods to approximate value function in RL using minimal domain knowledge and the general-purpose framework for representation learning (Sutton & Barto, 2018), and deep learning can learn the best representation directly from raw inputs. After the impressive performance of DQN, discovered by Mnih et al. (2013), DQN with target networks and double estimation to calculate Q-function are proposed to stabilize training and address the problem of overestimating (Mnih et al., 2015; Van Hasselt, Guez, & Silver, 2016). In this paper, we employ both modifications in our DRL method as Double Deep Q-Network with a target network (TN-DDQN).

4.1. Agent and environment

As alluded to previously, in our joint pricing and inventory problem with reference prices, the retailer stands in for the decision-making agent who learns to make the pricing and ordering decision, as shown in Fig. 2. In each step of the TN-DDQN algorithm, the retailer chooses an action based on observation of the environment, which is expressed by two-dimensional inputs: initial inventory level x_t, and customer's reference prices, r_t. The discrete action space is described as the same two-dimensional spaces constructed by order-up-to inventory level y_t, and the retailer's sales price, p_t. At the beginning of the infinite horizon, the observations the retailer receives are set to arbitrary values on account of the favorable universality and expansibility of the TN-DDQN algorithm. The inventory transition function is computed by the relation between order-up-to inventory level and stochastic demand in the current period. In addition, the exponentially smoothed adaptive expectations model is applied to the reference prices transition function. Finally, the profits encompassing sales revenue, procurement cost, and inventory cost in each period are collected as rewards.

The sales price gives the retailer leverage to adjust the everyday inventory level and behaves much more effectively with the help of reference prices. On the surface, this is a maximization problem for the retailer, but in reality, this is a minimization of inventory costs under infinite horizon.

4.2. Invalid action mask

When the retailer applies RL algorithms to interact with real-world scenarios like joint pricing and inventory problem, sometimes not all pricing and ordering actions are reasonable (or allowable) in particular states. If the retailer attempts redundant or irrelevant actions at period t, computation time could be wasted and the market environment may output incorrect rewards at the current period. A technique known as "invalid action masking" has been applied in recent work (Berner et al., 2019; Zahavy, Haroush, Merlis, Mankowitz, & Mannor, 2018), which masks out invalid actions for some particular states in the environment instead of setting extremely negative rewards for the invalid actions. Based on the technique of invalid action masking, we reform the action spaces for the retailer as shown in Fig. 3.

At each period, t, the order-up-to level, y_t must be more than the initial inventory level, x_t and the sales price, p_t, must be in the interval $[c_t, p_{max}]$ regardless of the customer's reference price, r_t. As a matter of fact, the process of invalid action masking can be regarded as a state-dependent (inputs of the DNN) function for discrete action spaces (outputs of the DNN).

4.3. Algorithm description

A DQN updates the DNN such as Convolutional Neural Networks (CNNs) to estimate $Q(s, a)$, and in our proposed algorithm with TN-DDQN, there are two independent DNNs with the same network structure but disparate weights and biases parameters, named "evaluated Q-Network" and "target Q-Network" respectively. The target Q-Network calculates the target Q value corresponding to the next state in Equation (6) and updates parameters with the evaluated Q-Network after a fixed number of steps. Remarkably, we reduce the overestimation of Q values by learning two Q-function estimates using different experiences. The Q maximizing action is selected by using the evaluated Q-Network, and the Q value used to calculate the target Q value is generated by the target Q-Network. Then, the evaluated Q-Network brings about its Q value by the current state and is trained by minimizing the loss between target Q value and its Q value through mean square error, as shown in Equation (7).

$$y^t = R^{t+1} + \gamma * Q'(X^{t+1}, \underset{A'}{max} Q(X^{t+1}, A'; \theta); \varphi) \tag{6}$$

$$L(\theta) = \mathrm{E}[(y^t - Q(X^t, A^t; \theta))^2] \tag{7}$$

Good policies should preserve a balance between exploring the action spaces and exploiting knowledge learned by the agent. We choose ε - greedy policies to balance exploration and exploitation by Equation (8):

$$\varepsilon = \varepsilon_{min} + (\varepsilon_{max} - \varepsilon_{min}) e^{-\varepsilon_{decay} * s} \tag{8}$$

where the exploration rate, ε, decreases exponentially with decay rate, ε_{decay}, from ε_{max} to ε_{min} as training episodes, s, increase, as depicted in Fig. 4.

Off-policy algorithms such as TN-DDQN do not have to discard experiences once they have been used and we can store the historical experiences in experience replay memory. Experiences are picked randomly from the replay memory so that highly correlated experiences can be avoided. Besides, experience replay could shorten the trial-and-error process (Lin, 1992) by facilitating the reuse of experiences. We

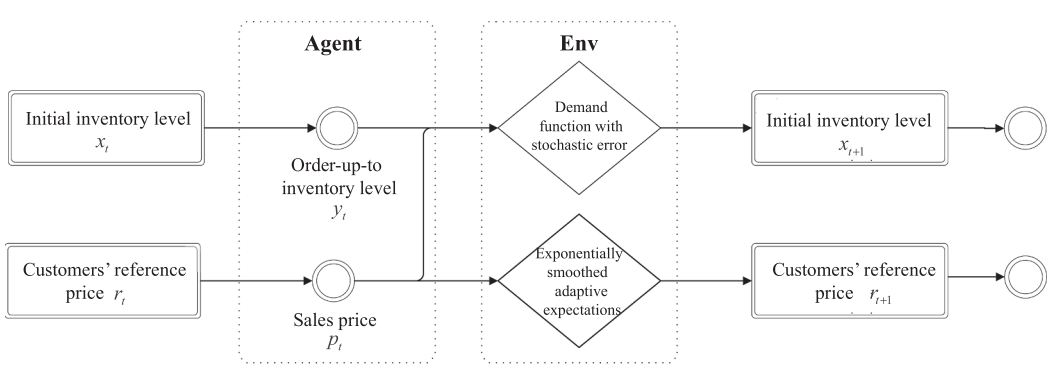

Fig. 2. Schematic diagram of agent and environment in TN-DDQN.

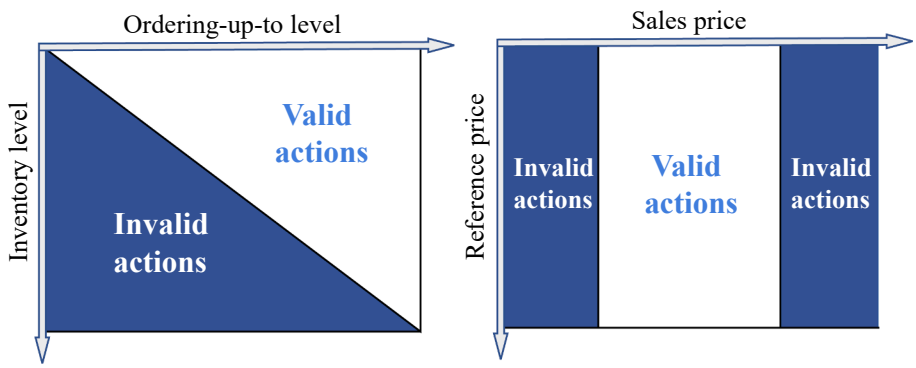

Fig. 3. Invalid action mask for the retailer.

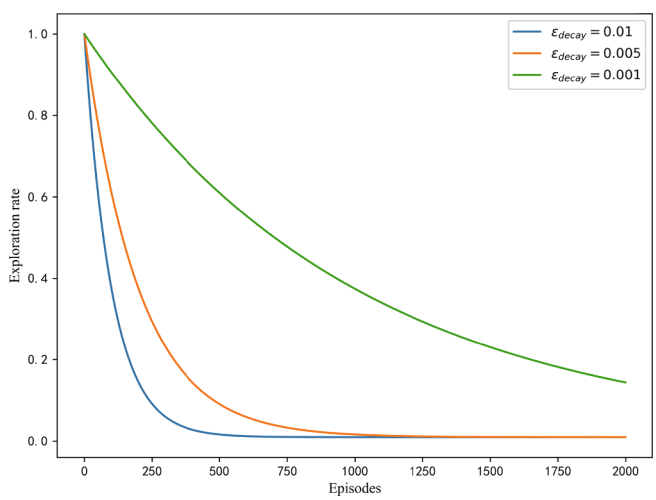

Fig. 4. Exploration rate ε with different ε_{decay}.

summarize our DRL method with TN-DDQN in **Algorithm 1**.

Algorithm 1 TN-DDQN for joint pricing and inventory problem with reference prices.

Initialize learning rate δ
Initialize target network update frequency C
Initialize experience replay memory with max size N
Initialize batch size B
Initialize exploration rate interval $[\varepsilon_{min}, \varepsilon_{max}]$
Initialize exploration rate decay $\varepsilon_{\text{decay}}$
Randomly initialize the Q network with parameters θ
Initialize the target Q^{target} network with parameters $\varphi = \theta$
For *episode* $= 1 : epochs$ **do**
Reset the marketing environment and initialize state X^0
Update exploration rate ε
For t $= 1 : T$ **do**
With probability ε, select a random valid action A^t, otherwise select $A^t = arg\ max_{A'} Q(X^t, A^t; \theta)$ in valid action space
Execute action A^t and observe reward R^{t+1} and X^{t+1}
Gather and store transition $(X^t, A^t, R^{t+1}, X^{t+1})$ in the experience replay memory
Set $X^{t+1} = X^t$
Sample a minibatch B of experiences from the experience replay memory
Calculate the target values using double networks by:
$y^t = R^{t+1} + \gamma * Q^{target}(X^{t+1}, max_{A'} Q(X^{t+1}, A'; \theta); \varphi)$
Calculate the loss using MSE by:
$L(\theta) = E_B[(y^t - Q(X^t, A^t; \theta))^2]$
Update the network's parameters by
$\theta = \theta - \delta * \nabla_\theta L(\theta)$
Every C steps update the target network by:
$\varphi = \theta$
End for
End for

5. Numerical experiments

We conduct numerical experiments to evaluate the performance of our proposed DRL algorithm and to reveal the significance of dynamic pricing involving long-term profits for infinite horizon inventory systems. The discrete and continuous state spaces are solved using DP, Real-Demand-Response, and TN-DDQN algorithms. In the numerical experiments, the expected mean episode profits are considered a measure of performance. The heterogeneity in customer valuation and other key parameters are also discussed in the situation with discrete state space.

Our DNN is designed as a fully connected network with two hidden layers, with each layer having 256 neurons and each node a rectified linear activation function. There is one output node for each possible value of the pricing and inventory decisions, and none of *meta*-information will be lost because of the independence and non-planarization of pricing and inventory actions. To optimize the proposed DNN network, we adopt the Adam optimizer (Kingma & Ba, 2014), which dynamically adjusts the step size in conjunction with previous updates. We train each agent on 5000 episodes, and in each episode, the agent proceeds with a 100-period simulation to approximate the environment of infinite-horizon MDP.

Indeed, determining the suitable hyper-parameters of the TN-DDQN algorithm and obtaining superior performance is a difficult task and formidable computation time is needed. To overcome the difficulty, we start from scratch and build several tips that work very well, rather than solely search the parameters in a grid directly when we train and tune the hyper-parameters of TN-DDQN.

First, the learning rate should be typically small enough to learn a more optimal or even global optimal set of weights. Conversely, the decay rate of greedy policies must be quite large, which means the exploration rate decreases quickly as time goes by and avoids sacrificing reward gathering. After determining the best-performing learning rate and decay rate, it is better to choose a small number of elements in a batch of experiences, so that each experience is likely to be sampled more than once, making learning more efficient. However, the larger size of experience replay memory and faster network-update frequency are unimportant, as these improve already well-performing models indistinctively.

The values of various parameters used in our experiment are presented in Table 3. For simplicity, the actions of pricing and ordering are restricted to discrete values and the probability function for the discrete random variable, σ, gives [-5, −4, −3, −2, −1, 0, 1, 2, 3, 4, 5] with probability distribution [0.02, 0.04, 0.08, 0.12, 0.14, 0.2, 0.14, 0.12, 0.08, 0.04, 0.02], respectively. The formation of demand function is:

$$d(p, r) = (a - bp) + \eta^{+} max\{0, (r - p)\} + \eta^{-} min\{0, (r - p)\} + \sigma \tag{9}$$

where a represents the market size, b weighs up the sensitivity of demand function with respect to the sales price, and η^{+} and η^{-} are reference price effects on demand.

Table 3
The basic parameter values for our numerical experiment.

Parameter	Value	Explanation
a	30	Marketing size
b	1	Scales of sensitivity of demand towards the sales price
c	1	Unit ordering cost
γ	0.9	Discount factor
α	0.5	Memory factors
η^+	0.5	Parameters of reference price effects when $r_t \geqslant p_t$
η^-	0.5	Parameters of reference price effects when $r_t \leqslant p_t$
h	6	Unit holding cost
v	3	Unit backlogging cost
p_{min}	10	The lower bound of sales prices and reference prices
p_{max}	20	The upper bound of sales prices and reference prices
ε_{max}	1	Maximum exploration rate
ε_{min}	0.01	Minimum exploration rate
ε_{decay}	0.005	Decay rate
B	32	Batch size
δ	0.0001	Learning rate
C	100	Target network update frequency
N	10,000	Experience replay memory

Following, all experiments are coded in Pytorch (version 1.9.0), and the results are obtained from a personal computer with a i7-8550U 1.99 GHz CPU and 8.00 GB of RAM.

5.1. Algorithm evaluation

We show the performance of the TN-DDQN algorithm on a joint pricing and inventory problem with reference prices in a numerical experiment.

We first conduct an MDP with discrete state spaces in which the agent (retailer) observes the discrete initial inventory level and the reference price. To solve the MDP, we employ value iteration (Bertsekas, 2000; Sutton & Barto, 2018) with a discount factor $\gamma = 0.9$. The value iteration algorithm is executed simply by turning the Bellman optimality equation into a particular update operation:

$$V_{k+1}(s) = \max_a \sum_{s',r} p(s', r|s, a)[r + \gamma V_k(s')], \ \forall s \in S \tag{10}$$

For arbitrary V_0, the sequence $\{V_k\}$ must converge to an optimal value function V^* in a finite number of iterations and the optimal policy can be obtained by the max operator. So, we compare the TN-DDQN algorithm with the optimal value iteration under the stochastic environment, as shown in Fig. 5. After 1500 training episodes, the TN-DDQN algorithm achieves stable conditions with slight fluctuations. The expected mean episode profits of value iteration and TN-DDQN are 198.33 and 197.21 respectively, which means our proposed algorithm can reach 99.43% of the best expected mean episode profits.

We evaluate the performance of the TN-DDQN algorithm in continuous spaces through real-demand circumstances. In particular, we set up an upper bound benchmark with real-demand-response (RDR) policy, in which the pricing policy is in accordance with our algorithm and its order quantity is always tantamount to the real demand in each period:

$$Price_t^{RDR} = Price_t^{TN-DDQN}, \ t = 1, 2, \ldots \tag{11}$$

$$Order_t^{RDR} = \overline{D}_t, \ t = 1, 2, \ldots \tag{12}$$

where $\overline{D}_t$ is the real demand in period t. Therefore, there is never any holding cost or backlogging cost for the entire horizon, which could be an objective metric to gauge the performance of the TN-DDQN algorithm in continuous spaces. Fig. 6 shows the gaps between the Real-Demand-Response policy and our TN-DDQN algorithm after 5000 simulations. The expected mean episode profits of Real-Demand-Response policy and TN-DDQN are 202.84 and 199.37 respectively, which means our amenable algorithm can reach 98.29% of the benchmark generally. Notably, the gaps of mean episode profits between discrete state spaces and continuous state spaces are mainly because of discretizing continuous values and mapping a range of infinite discrete values into a set of finite discrete values.

5.2. Comparison between (Double) Q-Learning and TN-DDQN

In this section, we compare (Double) Q-learning and the TN-DDQN algorithm in discrete state spaces and employ parameter analysis of backorder cost and holding cost.

Q-learning (Watkins & Dayan, 1992) is one of the powerful RL algorithms, and has been known to discover near-optimal solutions as a special case of value iteration. As an off-policy temporal-difference (TD) algorithm, the Q-learning algorithm enables the agent to learn about choosing the action leading to the maximum Q-value with an exploration rate in a greedy policy, and after each iteration the corresponding state-action pair is updated in the following manner:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \zeta[r_{t+1} + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t)] \tag{13}$$

where $\zeta \in (0, 1]$ is the step-size parameter, which decreases in the training process to assure convergence.

However, the use of the max operator to determine the next-state Q-

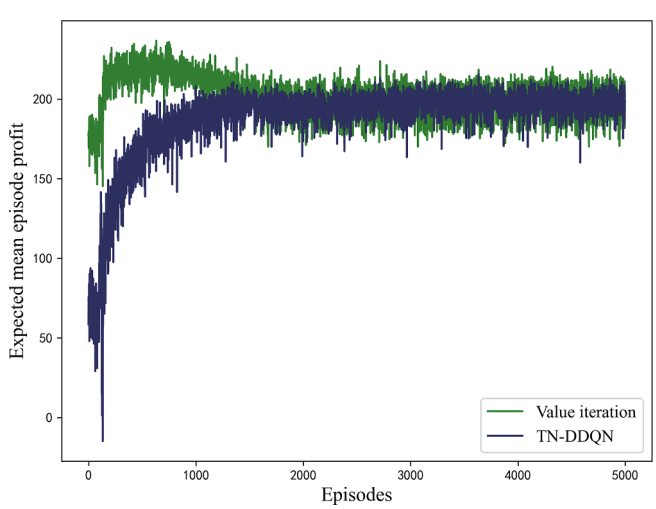

Fig. 5. Algorithm evaluation for TN-DDQN and value iteration.

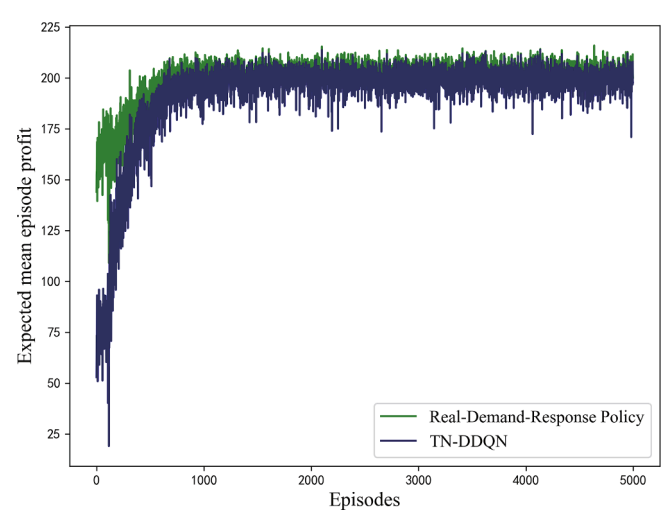

Fig. 6. Algorithm evaluation for TN-DDQN and Real-Demand-Response policy.

value can result in huge overestimations of the action values. Therefore, an alternative double estimator method (Double Q-learning) is proposed (Hasselt, 2010) and the update of TD in Double Q-learning is:

$$Q^1(S_t,A_t) \leftarrow Q^1(S_t,A_t) + \zeta[r_{t+1} + \gamma Q^2(S_{t+1},a^*) - Q^1(S_t,A_t)] \tag{14}$$

where $a^* = argmax_a Q^1(S_{t+1}, a)$, and Q^1, Q^2 are two independent Q-functions, which should be updated alternately or randomly.

First, we check out different parameters of step sizes and exploration rates in the form of piecewise monotone function, and over 3000 episodes, the parameters with best performances are chosen for Q-learning and Double Q-learning, respectively. After that, we implement a comparative experiment and present the results in Table 4 and Table 5.

Table 4 shows that the higher backorder cost and holding cost are, the bigger the gaps between (Double) Q-learning and TN-DDQN algorithm will become. In other words, our algorithm performs very well compared with the two classical RL algorithms, especially when the cost of "making mistakes" is high. Moreover, the Double Q-learning algorithm achieves higher average episode profits than the Q-learning algorithm by underestimating the maximum expected value. To explain in detail why the two RL algorithms' realizations are not outstanding and why the Q-learning algorithm performs worse than Double Q-learning, we calculate and record the average optima of order-up-to level, pricing and Q-function with TN-DDQN and (Double) Q-learning, as shown in Table 5. First, the average optimal order-up-to levels and the average pricing optima are essentially increasing with the unit backorder cost and decreasing with the unit holding cost for both algorithms. That is because when the unit backorder cost is very high, the retailer has an incentive to order more products and raise their prices to reduce the current demand, and vice versa. Of course, Q-functions decrease with both the unit backorder cost and unit holding cost. Second, the Q-function of Double Q-learning is smaller than that of Q-learning by reason of the double estimator method. Simultaneously, Double Q-learning has a lower optimal order-up-to level and pricing optima, which is closer to the optimal decisions compared with the Q-learning algorithm. Further, the loss of episode profits in Double Q-learning and Q-learning algorithms results from ordering too many products and setting too high sales prices.

5.3. Myopic pricing and inventory policy

Suppose there is an agent (retailer) who always makes the optimal myopic policy in a single-period, which maximizes the profits of the current period without considering the impact in long-term decisions. If customers are all loss-neutral, given the reference price and initial inventory level, it is always strictly concave in pricing and ordering decisions for the expected single-period revenue. The optimal myopic order-up-to level and pricing policy is calculated in Equation (15) and (16). Further, single-period revenue is strictly concave in the reference price and sales price if, and only if, customers are all loss-neutral.

$$y_m^* = \begin{cases} 0 \ if \ x \leqslant 0 \\ x \ else \end{cases} \tag{15}$$

$$p_m^* = \frac{a + \eta r + \nu(b+\eta)}{2(b+\eta)} \tag{16}$$

where η represents that the customers are loss-neutral.

We compute the expected mean episode profits of our TN-DDQN algorithm and myopic pricing policy when the discount factor and unit ordering cost both change arbitrarily. Table 6 shows that if the unit ordering cost or the discount factor in the environment increases, the loss due to myopic strategies will increase accordingly, which is intuitive. The results reveal retailers should consider the short-term and long-term benefits in a balanced way, because retailers can affect customers' next-period reference prices and their profits in the future through the current sales price.

5.4. Customer preference

In this section, we design two experiments to investigate customers' attitudes toward the relationship between reference prices and retailer's sales price.

One experiment relies on customer mental accounts (Thaler, 1990) in different kinds of sensitivity to gains and losses, named loss-aversion, gain-seeking and loss-neutral. We assume the parameters of gain and loss are the Cartesian product $N^{+}*N^{+}$, where N^{+} is discrete number set [0.1, 0.2, 0.3, 0.4, 0.5]. Then we launch 5000 episodes 25 times. The experiment's results, shown in Fig. 7, show the parameters of loss, η^{-}, and the proportion of TN-DDQN to value iteration are positively correlated. This means that the more loss-averse customers there are in the market which is consistent with prospect theory (Kahneman & Tversky, 1979), the better the performance our TN-DDQN algorithm achieves.

The second experiment is used to measure the impact of customers' memory factor. We implement experiments and show the average optima of pricing decision and order-up-to level in Fig. 8 with changes of the memory factor. When the memory factor is extremely small, the current sales price makes a great difference to future reference prices, and the retailer has an incentive to obtain a higher future reference price by improving the current sales price, through which they will acquire larger gain effects in future periods. Conversely, when the memory factor is big, the current pricing decision has little influence on the future state of reference prices. The retailer selects a lower price to obtain the gain effects at present on instinct. The mutual exclusion of pricing and order-up-to level is exhibited clearly in Fig. 8.

5.5. Optimal pricing and inventory policy

Fig. 9 visualizes the optimal value function and optimal control of DP and TN-DDQN. The top of the figure indicates the value function of our TN-DDQN algorithm in the state of the initial inventory and the reference price has pertinent distribution with the value function of DP, which can further explain the near-optimal characteristic of TN-DDQN. The middle of the figure is the optimal pricing decision in the state space. Although the optimal pricing decisions of DP and TN-DDQN are different in detail, both retain the list-price-policy property that the optimal pricing is decreasing with the initial inventory level. The

Table 4
Average episode profits ± standard deviation with TN-DDQN, Q-learning and Double Q-learning.

Unit backorder cost	Unit holding cost	TN-DDQN	Q-learning	Double Q-learning	Q-learning/TN-DDQN	Double Q-learning/TN-DDQN
1	3	201.17 ± 1.47	144.52 ± 0.70	147.64 ± 1.29	71.8%	73.4%
1	5	200.79 ± 0.69	118.06 ± 0.34	123.40 ± 0.97	58.8%	61.5%
1	7	199.52 ± 0.71	92.73 ± 0.80	99.87 ± 0.76	46.5%	50.1%
3	3	199.63 ± 2.41	143.21 ± 4.08	146.05 ± 1.05	71.7%	73.2%
3	5	198.03 ± 0.92	116.13 ± 1.44	120.74 ± 1.08	58.6%	61.0%
3	7	196.19 ± 0.24	89.36 ± 0.66	97.00 ± 2.27	45.4%	49.4%
5	3	198.20 ± 0.95	139.18 ± 1.14	142.25 ± 1.35	70.2%	71.8%
5	5	195.94 ± 0.79	111.73 ± 0.64	117.52 ± 1.91	57.0%	60.0%
5	7	193.53 ± 0.51	85.56 ± 0.87	93.40 ± 0.85	44.2%	48.3%

Table 5
Average optimal order-up-to levels, pricing optima and Q functions with TN-DDQN, Q-learning and Double Q-learning.

		Average optimal order-up-to levels			Average pricing optima			Average Q functions		
Unit backorder cost	Unit holding cost	TN-DDQN	Q-learning	Double Q-learning	TN-DDQN	Q-learning	Double Q-learning	TN-DDQN	Q-learning	Double Q-learning
1	3	21.11	27.66	25.87	13.79	15.25	14.41	2080.33	2287.16	1827.63
1	5	21.10	27.45	25.85	12.74	15.48	14.30	2058.95	2370.58	1793.72
1	7	21.09	27.11	24.65	12.96	15.35	14.29	2014.85	2393.72	1774.02
3	3	23.35	28.92	28.21	13.69	15.46	14.58	2034.81	2217.43	1778.78
3	5	22.87	28.86	26.41	12.13	15.43	14.68	2007.18	2319.41	1739.07
3	7	22.18	28.13	25.35	13.17	15.53	14.36	1967.34	2341.73	1708.30
5	3	24.09	29.79	28.66	14.26	15.53	14.69	2015.22	2211.55	1753.88
5	5	23.56	28.96	26.75	13.38	15.81	14.72	1966.95	2284.13	1704.57
5	7	23.55	28.44	26.19	13.35	15.68	14.45	1951.62	2311.86	1661.13

Table 6
Expected mean episode profits with TN-DDQN and myopic policy.

Discount factor	Unit ordering cost	TN-DDQN	Myopic policy
0.6	1	200.56	195.35
0.6	2	183.04	180.96
0.6	3	166.78	163.41
0.7	1	200.53	192.66
0.7	2	186.92	177.83
0.7	3	173.06	162.06
0.8	1	202.02	185.63
0.8	2	188.28	172.51
0.8	3	174.21	162.05
0.9	1	201.03	178.92
0.9	2	186.30	168.11
0.9	3	172.95	158.58

optimal order-up-to decisions at the bottom of the figure have little discrepancy between DP and TN-DDQN. The pattern of higher initial inventory level complying with higher order-up-to level is consistent with the constraints on the ordering strategies.

5.6. Environmental disturbances

The emerging trends in fierce market competition and the increased availability of demand data have led to pivotal and unprecedented challenges for companies. An obvious question is how the agent (retailer) can deal with a crisis and make the optimal pricing and inventory policy under the changeable external market environment. In this section, we simulate three different kinds of market environment with random disturbances in Table 7.

After conducting the experiments in three market environments for 5000 episodes respectively, we summarize the results in Fig. 10. Regardless of how the market environment changes, the expected mean episode profits of TN-DDQN remain at the level of 200 and the proportions of TN-DDQN to DP are all above 99%, which depicts that our TN-DDQN algorithm is more robust to stochastic disturbance changes in the market environment.

5.7. Summary of experimental results and managerial insights

We evaluate the TN-DDQN algorithm with several benchmarks, and show that, through our designed algorithm, the expected every-period profits can be maximized to achieve a near-optimal level compared with the DP algorithm and RDR policy. Further, TD learning algorithms are usually applied in supply chain management (Stockheim et al., 2003; Chaharsooghi et al., 2008; Dogan & Güner, 2015; Kara & Dogan, 2018), such as Q-learning and Double Q-learning. Therefore, these two algorithms are compared with the TN-DDQN to demonstrate the efficiency and effectiveness of our algorithm. More specifically, the Q-learning and Double Q-learning algorithms can only achieve around 58.3% and 60.9% performance of that of our TN-DDQN algorithm respectively. We then investigate customers' preference of gain-loss parameters and mental-memory factors and show there is a specific regular pattern between customers' behaviors and expected everyday profits. Moreover, our TN-DDQN algorithm is still an entrenched method if the marketing environment changes unpredictably with stochastic disturbances.

In terms of theoretical implications, we have analyzed the detailed

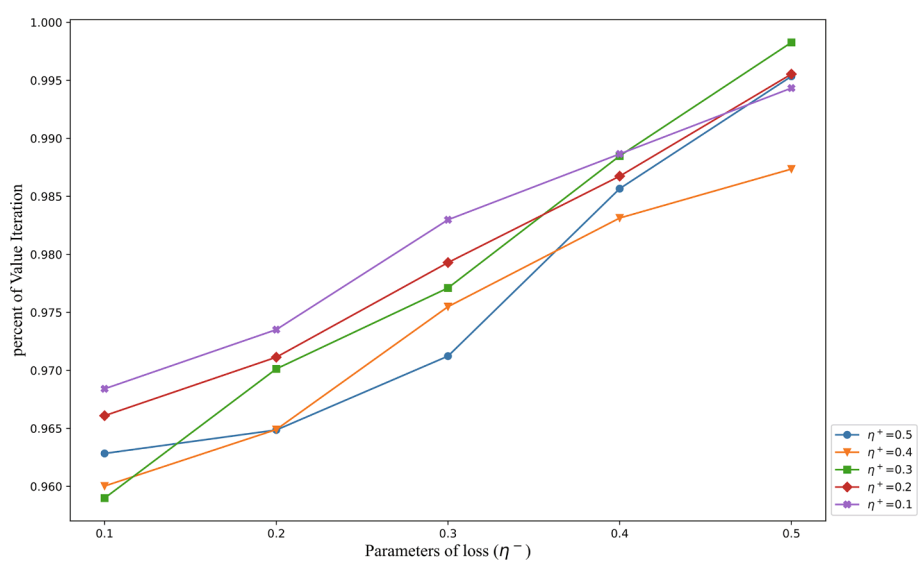

Fig. 7. Proportion of TN-DDQN to value iteration under the parameters of gain and loss.

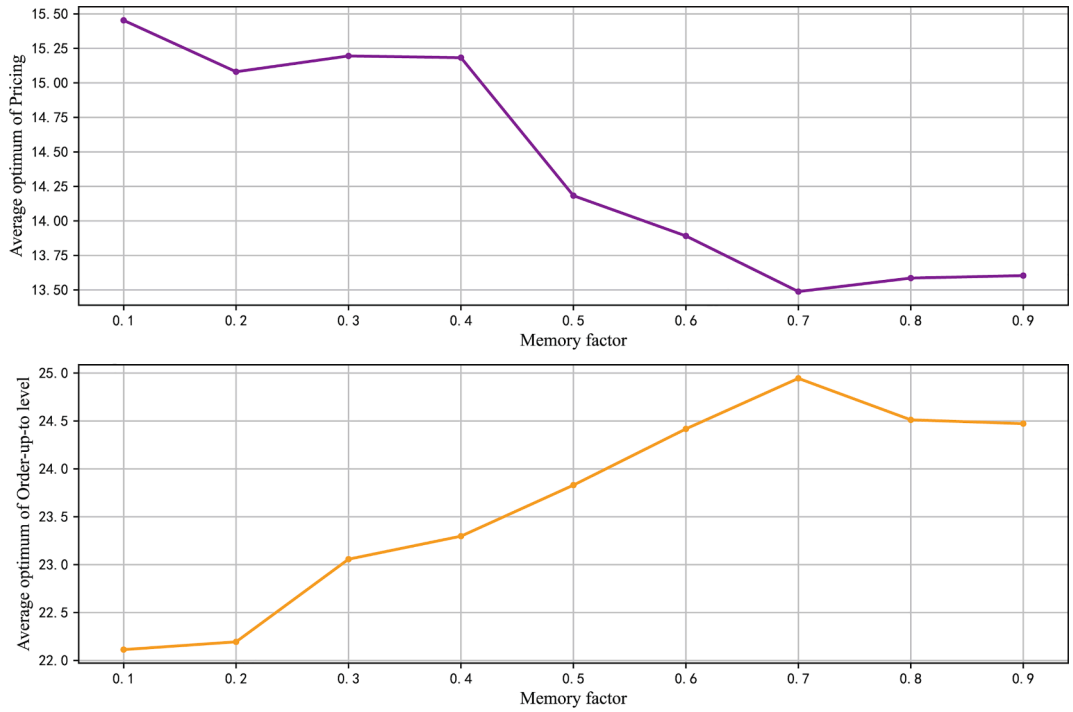

Fig. 8. Average optima of pricing and order-up-to level with the memory factor.

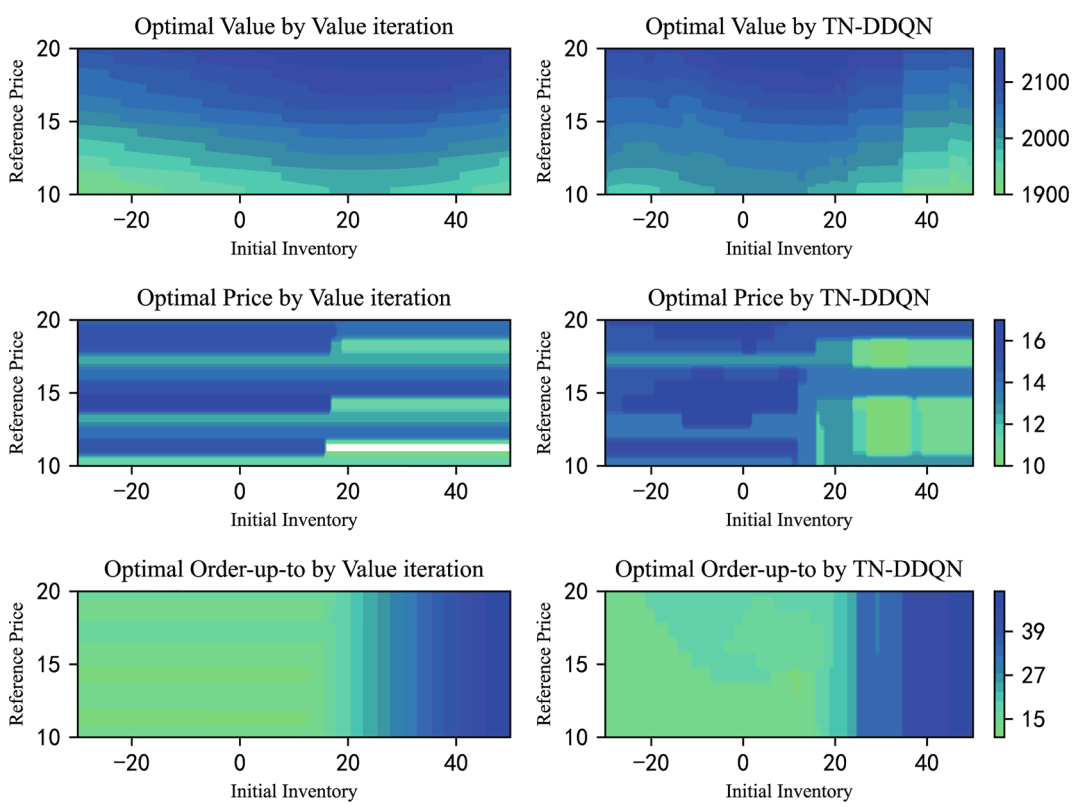

Fig. 9. Optimal value and optimal control between value iteration and TN-DDQN.

extensibility and feasibility of an invalid-action-mask mechanism for selections in action spaces to avoid estimation bias due to boundary effects. Additionally, our techniques of hyperparameter tuning are recommended theoretically and subtly exploit the advantages and traits of the TN-DDQN algorithm, instead of exploring the whole values defined for each hyperparameter. The obtained results of these experiments can also be discussed in terms of managerial insights. From the perspective of decision-makers, myopic policy inevitably causes retailers' profits to be lost, and these economic losses are magnified by discount factors and ordering costs. From the micro point of view, customers' ability to remember past prices forces retailers or companies into lowering sales prices and increasing the order-up-to level. It is therefore necessary to

Table 7
Three different kinds of market environment.

Market	The distribution of random disturbance	Standard deviation
1	[0.02, 0.04, 0.08, 0.12, 0.14, 0.2, 0.14, 0.12, 0.08, 0.04, 0.02]	2.227
2	[0.02, 0.04, 0.06, 0.08, 0.1, 0.4, 0.1, 0.08, 0.06, 0.04, 0.02]	2.049
3	[0.01, 0.02, 0.04, 0.06, 0.07, 0.6, 0.07, 0.06, 0.04, 0.02, 0.01]	1.574

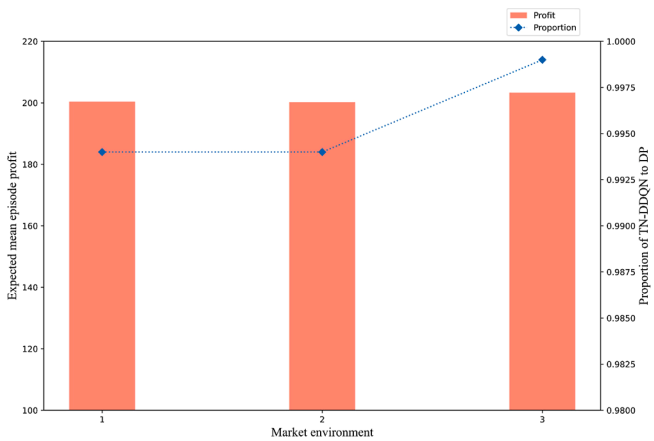

Fig. 10. Performance of TN-DDQN under environmental disturbances.

examine the latent impacts of reference prices under dynamic pricing and inventory control strategies.

6. Conclusion

In this study, we develop a joint pricing and inventory control as an infinite-horizon MDP, with reference prices based on the adaptation level theory predicting customers' responses to the sale price of an item by comparing it with an internal mental standard. Distinctive features of our study include fewer assumptions of customers' demand function and more generality of the objective value function. To cope with these difficulties, a deep RL algorithm based on Double Deep Q-Networks with a target network (TN-DDQN) is proposed by considering invalid action mask and the exploration versus exploitation trade-off. We demonstrate how the algorithm approaches the optimal DP and RDR policy in discrete and continuous state spaces respectively. Specifically, a rigorous performance evaluation of our algorithm compared with the tabular-based Q-learning and Double Q-learning algorithm, which is common in the RL literature, is investigated. When the decision-maker adopts optimal single-period pricing and ordering decisions, the optimal myopic policy achieves lower profits than policies from our algorithm. Moreover, it is also shown that if customers have high abilities to remember historical prices, the decision maker must bring sales prices down and increase the order-up-to level accordingly. Finally, with experiments of stochastic disturbances for a variety of marketing parameters, we show the promising robustness of the TN-DDQN algorithm, which can be extended to more realistic models of human behaviors.

Several extensions of our work are worthy of future investigation and development:

- The circumstances could be reconsidered with the assumption that the retailers and customers have bounded rationality with more complicated psychological factors, such as stochastic reference price effects or risk preference.
- Researchers could consider what happens to the modalities of pricing and inventory strategies as well as value function if there is a positive delayed time before products arrive at retailers under reference price effects.
- We focus on a monopolistic pricing and inventory control, whereas multiple agents are usually involved in supply chain management, and their reciprocities (cooperative, competitive or mixed relationships) may influence the total expected profits of multi-echelon supply chains, which is worth exploring.
- Different neural network architectures, such as long short-term memory or generative adversarial networks should also be explored for the purpose of increasing the performance in our proposed algorithmic framework.

CRediT authorship contribution statement

Qiang Zhou: Writing – original draft. **Yefei Yang:** Supervision, Writing – review & editing, Visualization. **Shaochuan Fu:** Resources, Formal analysis, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was partially supported by The Fundamental Research Funds for the Central Universities (Grant No.: 2020JBW002) and National Natural Science Foundation of China (Grant No.: 72102011).

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